

Birefringence-Free Lithium Niobate Waveguides

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Lithium niobate (LiNbO₃, LN) is widely used for photonic applications, thanks to its wide transparency range, electro-optic and nonlinear optical properties [1]. Recently available lithium-niobate-on-insulator (LNOI) substrates afford high index contrast and thus tight field confinement in LN photonic devices such as optical modulators and microring resonators [2, 3]. However, similar to other (e.g. SOI) platforms, high-confinement waveguides are strongly birefringent, which leads to e.g. polarization mode dispersion with deleterious effects for delay-based filters, modulators and other signal processing and transmission devices [4]. Here, with vectorial simulations for high-confinement ridge waveguides in LNOI, we show how the interplay of the high birefringence of LN and of the waveguide structure can be employed to achieve birefringence-free designs in a broad spectral range, from 0.8 up to 1.8 μm , through a careful tuning of the LNOI ridge dimensions.

Fig. 1(a) illustrates the concept with the results of numerical simulations made for 300 nm-thick Y-cut LNOI ridge waveguides buried in SiO₂ (see inset) with varying ridge widths (w). The plot shows the computed effective indices for several TE and TM modes as a function of the waveguide width, for a given wavelength ($\lambda = 1064$ nm). It highlights the existence of multiple crossing points, where the indices of different mode pairs can be equalized through a suitable choice of the waveguide width, with the lowest order (TM_{00} - TE_{00}) crossing point occurring for $w = 398$ nm.

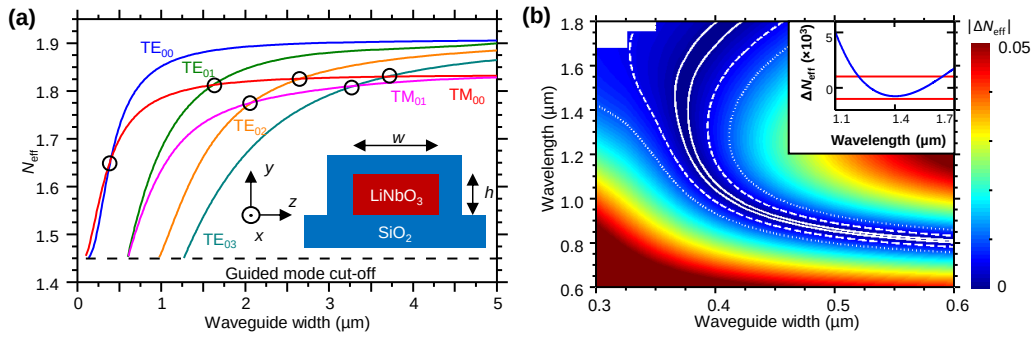


Fig.1 (a) Effective indices (N_{eff}) of the first six guided modes calculated at the wavelength $\lambda = 1064$ nm, for silica-clad buried LN waveguides, as a function of the waveguide width (w). The circles highlight the points where two guided modes experience the same effective index. The inset shows the waveguide cross section. The axes represent the LN crystal axes. (b) Contour plots of the TM_{00} to TE_{00} mode birefringence $|\Delta N_{\text{eff}}|$ as a function of waveguide width and wavelength. The solid, dashed and dotted contour lines represent $|\Delta N_{\text{eff}}| = 0.001$, 0.005 and 0.01 respectively. The white area corresponds to the guided mode cut-off. The inset shows $|\Delta N_{\text{eff}}|$ as a function of wavelength for a fixed waveguide width, $w = 376$ nm.

Fig.1 (b) further investigates the TE_{00} - TM_{00} mode birefringence ($|\Delta N_{\text{eff}}|$) as a function of wavelength (λ) and waveguide width (w). From the contour plots of $|\Delta N_{\text{eff}}|$ in Fig. 1(b) it is apparent that zero-birefringence operation can be achieved in a broad spectral range, (from 800 nm up to the guided-mode cut-off around 1.8 μm), through a suitable choice of the waveguide width ($w = 350$ -600 nm). Furthermore, for a fixed waveguide width ($w \sim 376$ nm), close to the turning point in the 2D map of Fig. 1(b)), simulations predict birefringence-free operation within a $|\Delta N_{\text{eff}}| \leq 0.001$ tolerance in a broadband wavelength range ($\lambda \sim 1.22$ to 1.64 μm).

In conclusion we show that it is possible to engineer LN photonic wires so that the lowest order modes experience the same indices in a broad wavelength range in the infrared, encompassing also the telecom band. The required waveguide widths are in the 0.35 to 0.6 μm range and appear fully feasible in light of the most recent developments of integrated optics technology on LNOI [5]. Furthermore, these results can pave the way to implementing advanced polarization control and new functionalities in integrated and electro-optic devices, micro-resonators, and nonlinear frequency converters, coming of age now on LNOI.

References

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